



From “Lemon markets” to digital trust: How blockchain reshapes carbon trading and incentivizes cleaner production—evidence from agent-based simulation and global case studies

Lei Lei^a, Fei Xue^{b,*} , Jinxin Zhang^{a,c}, Malan Huang^{a,c}, Caibo Liu^{a,d}

^a Business School, Hubei University, Wuhan, 430062, China

^b China Three Gorges University Engineering Research Center of Eco-environment in Three Gorges Reservoir Region Ministry of Education, Yichang, 443002, China

^c Research Center for China Agriculture Carbon Emission Reduction and Carbon Trading, Hubei University, Wuhan, 430062, China

^d Wuhan College, Wuhan, 430212, China

ARTICLE INFO

Keywords:

Blockchain
Carbon markets
Information asymmetry
Digital MRV
Agent-based modeling
Cleaner production

ABSTRACT

Carbon markets are supposed to connect financial incentives with real-world decarbonization. In practice, that connection is often fragile. Credit quality varies, verification is costly, and supply-chain emissions are notoriously hard to observe. This paper asks whether blockchain-based infrastructures can ease some of these frictions, and under what conditions they might do so. To answer that question, the paper develops a Technology–Market–Production (TMP) framework linking digital verification architecture, carbon-market microstructure, and production-side decarbonization incentives. Empirically, it combines a mechanism-oriented agent-based model (ABM) with a comparative assessment of 16 blockchain-related carbon-market and industrial traceability initiatives.

The ABM is used as a computational laboratory rather than a tool for real-world causal identification. It examines whether lower verification frictions, under different institutional assumptions, can alter liquidity and price formation. The case analysis serves a different purpose. It is used to assess whether real-world institutional arrangements are broadly consistent with, complicate, or qualify the mechanisms proposed by the TMP framework.

Three findings stand out. First, digital infrastructures that reduce verification frictions may improve market functioning, but the scale and stability of those gains remain highly conditional. Second, more liquidity does not necessarily produce better environmental outcomes. When tokenized carbon assets remain loosely connected to underlying physical processes, greater tradability can coexist with persistent quality uncertainty. Third, systems with stronger physical anchoring—especially those linked to digital MRV and supply-chain traceability—appear more relevant to cleaner production than purely exchange-oriented models, although the available evidence remains uneven. Overall, the paper argues that credible digital reform in carbon markets depends less on blockchain alone than on how settlement, verification, and source-level data integrity are governed together.

* Corresponding author.

E-mail address: xuefei@ctgu.edu.cn (F. Xue).

1. Introduction

Carbon markets are built on a simple policy intuition. Put a price on emissions, and that price should begin to shape behavior. Firms that can reduce emissions at lower cost should have an incentive to do so. Capital, at least in theory, should move toward cleaner technologies and lower-carbon production. Yet real carbon markets rarely operate under those tidy conditions. Verification is expensive. Information is incomplete. Asset quality is difficult to judge. In some parts of the market—especially where supply-chain emissions are involved—the distance between the traded claim and the underlying physical reality can become uncomfortably large.

That gap is not incidental. A carbon credit is not a standard financial claim whose value can be checked through a cash flow or a balance sheet entry. Its quality depends on whether the underlying emissions reduction is additional, durable, non-duplicative, and properly recorded. These are not qualities buyers can easily verify on their own. Once quality becomes difficult to observe, the familiar problem of adverse selection appears. Lower-integrity credits can circulate beside higher-quality ones, and the market may struggle to tell the difference. In that setting, more trading does not automatically mean a better market.

This is one reason blockchain and related distributed ledger technologies have drawn so much interest in environmental markets. The appeal is easy to see. Distributed ledgers may strengthen traceability. Smart contracts may reduce settlement frictions. Tokenization may broaden access and make carbon assets easier to exchange. More broadly, blockchain is often described as a way of moving from institution-heavy trust toward a more auditable and programmable transaction environment. That sounds especially attractive in a market long troubled by duplicate claims, poor transparency, and fragmented registries.

Still, the economic significance of this shift is not self-evident. A ledger can preserve what has been entered into the system, but it cannot by itself guarantee that the underlying data were accurate when they entered. That distinction matters a great deal in carbon markets. If tokenization expands circulation faster than verification improves, the result may be a more liquid market for claims whose environmental integrity remains uncertain. In other words, a digitized carbon market can still look very much like a lemon market.

A second issue runs deeper. Much of the current discussion treats digital carbon markets either as a financial innovation problem or as a traceability problem. Less attention is paid to the link between the two. Even if blockchain-based systems improve market access and reduce transaction frictions, why should that change production decisions? Under what conditions would better market infrastructure translate into stronger incentives for firms to undertake verifiable decarbonization, especially in supply chains where Scope 3 emissions remain difficult to monitor?

This paper takes up that question by developing a Technology–Market–Production (TMP) framework. The framework links three layers that are often discussed separately. The first is the technology layer, where digital verification architecture affects information quality and auditability. The second is the market layer, where that information environment shapes liquidity, execution, and quality differentiation. The third is the production layer, where firms respond to the resulting financial and informational signals. The argument is not that blockchain automatically improves carbon markets. It is more conditional than that. Blockchain-based infrastructures matter economically only when they change the way information moves from verification systems into prices, and from prices into production incentives.

That means the paper is not making a blanket pro-technology claim. Digital infrastructures may reduce some verification frictions and improve parts of market functioning. But those gains depend on institutional design. A system that makes assets easier to trade without improving the credibility of the underlying data will not solve the deeper information problem. By contrast, systems that are more tightly anchored to physical processes—particularly through digital MRV and supply-chain traceability—are more likely to matter for cleaner production.

Methodologically, the paper combines two types of evidence. First, it uses an agent-based model (ABM) as a computational laboratory. The model is not asked to estimate a real-world treatment effect. It is used more modestly, to examine whether lower verification frictions can plausibly alter liquidity and price formation under stylized assumptions. Second, the paper compares 16 blockchain-related carbon-market and industrial traceability initiatives. These cases are not presented as causal proof either. They are used to assess where the proposed mechanism appears to fit real institutional settings, where it does not, and where it remains incomplete.

The paper makes three contributions. First, it offers a more integrated way of thinking about digital carbon markets by linking verification architecture, market microstructure, and production-side incentives within one analytical frame. Second, it uses simulation with some restraint. The ABM is not made to do more than it reasonably can. Third, it shows that liquidity cannot carry the whole argument. A carbon market can become easier to trade without becoming more informative in any meaningful environmental sense.

The rest of the paper proceeds as follows. Section 2 reviews the relevant literature and develops the TMP framework. Section 3 explains the research design, including the ABM, parameter discipline, and comparative case strategy. Section 4 presents the simulation and comparative findings. Section 5 discusses what those findings imply for digital carbon-market design. Section 6 concludes.

2. Literature review and theoretical framework

2.1. Information asymmetry, verification frictions, and carbon-market quality

The basic problem in carbon markets is easy to describe even if it is difficult to solve. Quality is hard to observe. Buyers often cannot verify whether a carbon credit reflects a genuine, additional, and durable emissions reduction. They rely instead on systems of measurement, reporting, verification, and registry governance. When those systems are costly or uneven, the market inherits a classic information problem. In that sense, carbon credits resemble credence goods. Once quality becomes difficult to distinguish, lower-quality assets can crowd out better ones, and prices lose much of their informative value (Akerlof, 1970).

This matters for environmental economics more broadly. Market-based instruments are often expected to internalize environmental externalities and guide firms toward cleaner production (Coase, 1960). But that logic depends on credible information. When monitoring is weak and transaction costs are high, the link between price signals and actual environmental outcomes becomes fragile. Work on environmental policy has long shown that information frictions and institutional costs can weaken the effectiveness of market-based mechanisms (Jaffe et al., 2005; Stiglitz, 2002). Carbon markets therefore do not fail only because prices are too low or regulation is incomplete. They also fail when institutions cannot reliably distinguish high-integrity assets from low-integrity ones.

The problem becomes sharper as emissions accounting moves beyond direct emissions and into supply chains. Scope 3 emissions are difficult to observe not only because the data are dispersed, but because reporting often depends on multiple firms with different incentives, standards, and disclosure constraints. Data are fragmented. Reporting cycles are slow. Firms may be reluctant to share operational information that reveals sourcing, costs, or strategic relationships. The result is not just a measurement problem. It is also a coordination problem.

2.2. Blockchain, tokenization, and the limits of digital trust

Blockchain has entered this discussion as a possible institutional technology. The early technical and policy literature often focused on record integrity, transfer traceability, and the prevention of duplicate claims. Distributed ledgers, in principle, can improve the audit trail surrounding issuance, transfer, and retirement of carbon assets (Macinante, 2017; Schletz et al., 2020). From an economic perspective, the attraction lies less in the technology itself than in its possible effect on verification and coordination costs. If a digital system lowers the cost of checking asset histories, reconciling transactions, and restricting conflicting claims, it may alter market participation and exchange conditions. This logic is consistent with Catalini and Gans (2020), who argue that blockchain's economic value stems largely from lower verification and networking costs.

The financial literature adds another layer. Tokenization is often associated with broader participation, lower entry barriers, and deeper secondary-market liquidity (Howell et al., 2020; Parhamfar et al., 2024). In carbon markets, that possibility is appealing. A more accessible market could, at least in theory, bring in more buyers and sellers, reduce frictions, and improve price discovery.

Yet that is only part of the story. Carbon credits are not purely digital claims. Their environmental value still depends on off-chain facts. This is where the “oracle problem” becomes central. A distributed ledger can record what is entered. It cannot guarantee, on its own, that the underlying real-world data are accurate. Cong and He (2019) make this point clearly in a broader blockchain context. Applied to environmental markets, the implication is straightforward: a more liquid digital market may still trade claims whose physical basis remains weakly verified.

This distinction matters because liquidity and credibility are not the same thing. A market may become easier to enter, faster to clear, and more continuous in trading, yet still fail to improve quality differentiation. If tokenization reduces trading frictions faster than it improves verification, digitalization may amplify circulation without resolving the underlying adverse-selection problem.

2.3. From market architecture to production-side incentives

A further gap in the literature concerns the connection between digital market design and production-side behavior. Existing research often treats blockchain either as a market infrastructure or as a traceability tool. Fewer studies explain how these two aspects interact to shape firms' incentives to decarbonize.

That missing link matters. Cleaner production does not follow from liquidity alone. Firms respond to the broader informational environment in which carbon claims are evaluated. If emissions reductions can be measured more credibly, attributed more clearly, and rewarded more consistently, firms may face stronger incentives to undertake verifiable low-carbon adjustments. If not, then improvements in market infrastructure may stop at the trading layer.

This issue is especially relevant in supply chains. Scope 3 emissions remain difficult to observe because data are scattered across multiple entities and often embedded in operational systems not designed for carbon accounting. Digital MRV systems, sometimes combined with sensors, product-level tracking, or supply-chain traceability tools, are often presented as a way to narrow this gap. Their potential lies in improving data granularity and making emissions-related information more auditable. Recent work suggests that blockchain-related infrastructures can influence firms' low-carbon innovation strategies and information environments in this direction (Zhu et al., 2024). Other studies point to the use of blockchain-enabled traceability for carbon accounting in complex operational settings such as aviation and logistics (Patro et al., 2025).

Although Green Total Factor Productivity (GTFP) provides a useful long-run lens for thinking about cleaner production, it is not operationalized in this paper as an estimated dependent variable. In the present study, GTFP is retained only as a broader conceptual horizon for future empirical research, while the analysis focuses more narrowly on verification conditions, market functioning, and production-side decarbonization incentives.

2.4. The Technology–Market–Production (TMP) framework

The TMP framework developed here links these strands into a single transmission structure. It does not treat blockchain as a stand-alone solution. Nor does it use the term “digital trust” in a loose or celebratory way.

At the technology layer, the issue is whether digital verification architecture reduces uncertainty surrounding transaction-relevant information. This includes record consistency, auditability, and the cost of reconciling asset histories. In this paper, that effect is captured under the broad label of algorithmic trust. The term should be read carefully. It does not mean unconditional trust in code. It

means only that certain verification frictions may fall when digital systems are credibly designed and governed.

At the market layer, the question is how the information environment affects exchange. If verification becomes easier and asset histories become more transparent, buyers and sellers may differentiate more effectively across asset quality. Participation may increase. Transaction frictions may fall. Liquidity may deepen. But this mechanism is not automatic. If the system improves tradability without improving the informational basis of quality, then liquidity gains may coexist with persistent adverse selection.

At the production layer, the key question is whether the market now generates more credible signals for firms. Better traceability and clearer quality differentiation may make low-carbon action more visible and more economically meaningful. Under those conditions, firms may face stronger incentives to undertake verifiable decarbonization. But this link can also fail. If source-level data remain weak or manipulable, the market signal may still be too noisy to shape real production decisions.

The TMP framework therefore proposes a conditional chain:

Digital verification architecture→**Information quality and verification frictions**→
Liquidity and quality differentiation → **Cleaner production incentives**.

This framework would be weakened if digital transparency did not reduce practical verification frictions, if liquidity gains did not improve quality differentiation, if physically integrated digital MRV systems did not materially improve traceability, or if hybrid governance systems systematically underperformed simpler alternatives.

These conditions are not included merely as interpretive caveats. They define the observable patterns under which the TMP framework would be weakened or rejected. If digital transparency fails to reduce practical verification frictions, if liquidity gains do not improve quality differentiation, or if physically anchored systems do not yield more credible traceability than conventional arrangements, then the central transmission logic proposed in this paper would lose much of its explanatory force.

3. Methodology

3.1. Research design

This study combines a mechanism-oriented agent-based model with a comparative multiple-case analysis. The two components serve different purposes. The ABM is used to explore whether the mechanism proposed by the TMP framework is internally coherent under stylized assumptions. The case analysis is then used to see how that mechanism encounters institutional heterogeneity in practice. The design is therefore sequential and mutually informing, but it is not a strong confirmatory triangulation exercise.

The paper does not claim to estimate a real-world treatment effect of blockchain adoption. Nor does it claim that the cases provide definitive causal validation. The goal is more modest: to examine whether lower verification frictions can plausibly change market dynamics in a stylized environment, and whether real-world digital-carbon initiatives display patterns that broadly fit, complicate, or limit that mechanism.

3.2. Study 1: Agent-based model as a computational laboratory

3.2.1. Modeling objective

The ABM is designed to represent a simplified carbon-trading environment in which information quality, verification frictions, and execution conditions interact. The model is not meant to reproduce any specific carbon market in full institutional detail. It focuses on a narrower question: if verification-related frictions decline, under what conditions might market participation, liquidity, and price formation change?

An ABM is useful here because carbon markets involve heterogeneous agents, partial information, repeated interaction, and feedback between market conditions and trading decisions. A bottom-up simulation is therefore well suited to examining how local behavioral rules generate aggregate market patterns.

3.2.2. Agents and environment

The artificial market includes three types of agents:

- **Emitters, representing firms with heterogeneous compliance or offsetting needs;**
- **Verifiers, representing institutions or processes that affect verification quality and cost;**
- **Digital infrastructure providers, representing the availability of blockchain-enabled trading and verification architecture.**

The market environment is stylized. Carbon assets are exchanged through repeated interaction over time. The model does not attempt to reproduce the full legal and institutional complexity of either compliance or voluntary carbon markets. It captures only the mechanisms needed for the present argument.

3.2.3. Core mechanisms and formal expressions

The model includes three linked mechanisms.

- (1) Information quality and algorithmic trust

In the baseline setting, agents discount carbon assets because they are uncertain about quality. In the digital-infrastructure scenario, verification-related uncertainty falls because transaction-relevant information becomes easier to trace and reconcile. We represent the perceived quality of asset (i) at time (t) as:

$$[\tilde{q}_i, t = q_i + \theta B_t - \varepsilon_i, t \setminus \text{tag1}]$$

where $\tilde{q}_i$ is the latent environmental quality of the asset, indicates whether the digital infrastructure is active, is the transparency effect, and captures residual uncertainty. A larger θ implies that digital verification reduces the discount associated with quality opacity.

(2) Transaction-cost reduction

Digital systems may reduce settlement, reconciliation, and verification frictions. The effective transaction cost at time is written as:

$$[c_t = c_0(1 - \beta B_t) + \xi_t \setminus \text{tag2}]$$

where c_0 is the baseline transaction cost, β is the marginal cost-reduction coefficient associated with digital infrastructure, and captures idiosyncratic market frictions. A higher β implies that smart-contract execution and improved record consistency lower the expected cost of completing trades.

(3) Liquidity formation and order matching

The original version of the model treated order matching as a function of market depth alone. That specification was too narrow. In the revised model, execution depends jointly on market depth, spread-related frictions, and order-level price deviation. The probability of a successful match is defined as:

$$[P_t^{\text{match}} = \frac{1}{1 + \exp[-(\alpha \text{Depth}_t - \gamma \text{Spread}_t - \lambda |p_t^b - p_t^a|)]} \setminus \text{tag3}]$$

where Depth_t is market depth, Spread_t is the bid–ask spread or a spread proxy, $|p_t^b - p_t^a|$ is the absolute gap between bid and ask prices, and α, γ, λ govern the sensitivity of execution to each component. This remains a reduced-form specification, but it avoids implying that liquidity depends on depth alone.

3.2.4. Parameter discipline

The earlier version of the study overstated the strength of its calibration. The revised paper takes a more careful position. It does not claim formal empirical calibration from a standardized observational dataset. Such a dataset does not yet exist for the range of blockchain-based carbon-market initiatives considered here. Instead, the model uses parameter discipline.

For each key parameter, the paper specifies its economic meaning, the benchmark value used in the baseline simulation, the plausible range considered in sensitivity analysis, and the indicative basis used to constrain that range. These ranges are informed by project materials, market-design documents, and related research on blockchain, environmental markets, and traceability systems. This is weaker than full econometric calibration, and the limitation is acknowledged openly. A fuller rationale for the benchmark values and robustness ranges is reported in [Appendix Table A1](#).

3.2.5. Experimental design

The ABM compares a baseline market environment with one or more digital-infrastructure scenarios. The baseline reflects higher verification frictions, lower information transparency, and more cumbersome transaction processing. The digital-infrastructure scenarios alter that environment by lowering the relevant frictions.

The simulation runs over repeated periods. Multiple runs are conducted under different initial seeds to reduce dependence on any single starting condition. The main outcomes of interest are price dynamics, participation, liquidity-related conditions, and the extent to which quality uncertainty remains reflected in market behavior.

The interpretation is comparative rather than predictive. Any simulated premium or liquidity shift should be read as a model-dependent outcome under specified assumptions, not as an externally valid estimate.

3.2.6. Sensitivity analysis

Because the results depend on parameter choices, the revised analysis places more emphasis on sensitivity. Parameters governing verification frictions, transaction-cost reduction, and execution conditions are varied across plausible ranges. The point is not to search for a single “true” estimate. It is to examine whether the basic comparative pattern survives and where it weakens.

3.3. Study 2: Comparative multiple-case analysis

3.3.1. Purpose of the case analysis

The second part of the study examines 16 blockchain-related carbon-market platforms, registry systems, tokenization initiatives, and industrial traceability applications. The purpose is not to treat these cases as direct empirical proof of the TMP framework. Rather,

the cases are used to assess whether observed institutional arrangements display patterns that are broadly consistent with, contrary to, or informative for the mechanism developed in Section 2.

This distinction matters. The evidence available across cases is uneven in both form and strength. Some initiatives are documented mainly through official institutional releases or exchange materials, while others are more visible through project documentation, corporate sustainability reports, or publicly available technical materials. For that reason, the case analysis is interpretive and comparative rather than confirmatory in a narrow causal sense (Eisenhardt, 1989; Yin, 2018).

3.3.2. Case-selection strategy

The cases were selected through theoretical sampling rather than random sampling. Given the speed with which digital-carbon infrastructures evolve, random selection would risk over-representing obsolete proof-of-concept projects or under-specified pilot claims. Instead, the sampling strategy was designed to maximize variation along two dimensions central to the TMP framework: governance structure and degree of physical anchoring (Eisenhardt, 1989; Yin, 2018).

On the first dimension, the sample includes state-led or consortium-based infrastructures, regulated exchange-linked models, DeFi-native architectures, and industry-integrated systems. On the second, it spans projects oriented primarily toward liquidity and market access as well as projects tied more directly to traceability, product-level carbon accounting, or supply-chain emissions visibility.

Following the logic already established in the original manuscript, three screening principles were applied. First, projects had to show a degree of operational maturity beyond purely conceptual white-paper status. Second, they had to provide at least some identifiable evidence regarding market design, verification architecture, traceability, or implementation scope. Third, the sample needed to preserve variation across four broad institutional clusters: state-led models, RWA/institutional financial models, DeFi-native models, and industry-integrated models. This strategy improves analytical leverage, although it does not eliminate the possibility of survivorship bias or success-case overrepresentation (Yin, 2018).

3.3.3. Evidence categories and data construction

To avoid relying on a single source type, the case analysis draws on three categories of evidence.

The first is technical and institutional documentation. This includes official architecture papers, regulatory or policy releases, public project documentation, and infrastructure descriptions. Examples include HKMA materials on Project Ensemble, exchange and operating materials from Climate Impact X, public documentation surrounding regulated or quasi-regulated infrastructures, and institutional materials related to carbon registry design (HKMA, 2024; Schletz et al., 2020).

The second is market-operation and platform evidence. This includes exchange materials, platform disclosures, publicly available infrastructure documentation, and, where relevant, evidence on bridging, tokenization, or liquidity architecture. For tokenization-oriented models, this category helps establish whether a project primarily changes circulation, settlement, or access conditions rather than underlying verification quality (Howell et al., 2020; Parhamfar et al., 2024).

The third is operational and traceability-related evidence. For industry-integrated cases, this includes public sustainability disclosures, traceability-system materials, and project-level descriptions of upstream monitoring, battery passport design, supply-chain tracking, or data-sharing infrastructure. These materials are useful for identifying how digital systems are being used to structure visibility around Scope 3 emissions and operational carbon information, although they are not treated as independent causal proof of emissions reduction (Molinaro et al., 2026; Patro et al., 2025; Zhu et al., 2024).

These sources are not equally probative. Corporate sustainability reports, for instance, may be informative about implementation direction and disclosure priorities, but they do not carry the same evidentiary weight as formal registry documents or clearly specified institutional rules. Cross-case comparability is therefore imperfect across infrastructures, jurisdictions, and governance settings. The comparative case analysis should be read as an effort to identify institutional variation and recurring patterns, not to impose a single evidentiary standard across all projects.

3.3.4. Analytical logic and case grouping

The comparative analysis proceeds by asking three questions of each case.

First, does the initiative appear to reduce verification frictions, improve traceability, or alter the informational environment of carbon trading in a meaningful way? Second, does it primarily expand liquidity and tradability, or does it also strengthen quality differentiation and physical anchoring? Third, does its institutional design appear capable of transmitting digital improvements into stronger production-side decarbonization incentives?

To organize the comparison, the cases are grouped into four clusters, each corresponding to a different part of the TMP framework.

State-led models focus on settlement credibility, sovereign control, and registry discipline. Cases such as Project Ensemble, Climate Impact X in its institutionalized marketplace dimension, China Green Trust Chain, and the Zimbabwe registry initiative are relevant here because they illuminate how regulatory authority, settlement finality, and formal market architecture can shape trust conditions in carbon-market infrastructure (HKMA, 2024; Schletz et al., 2020).

RWA/institutional financial models emphasize tokenization, custody, and liquidity under more formal financial intermediation. Cases such as J.P. Morgan Kinexys, Northern Trust Carbon Ecosystem, ACX, and Thallo are included because they show how carbon-related claims can be integrated into broader digital-asset or regulated-market infrastructures. Their relevance lies mainly at the market layer of the TMP framework: they are useful for examining how carbon assets may become more tradable, more standardized, or more institutionally legible as financial instruments (Catalini & Gans, 2020; Howell et al., 2020).

DeFi-native models are included because they illuminate the opportunities and limits of permissionless digital carbon markets. Carbonmark, Senken, Toucan Protocol, and Regen Network differ in design and ambition, but together they make visible the tension

between access, liquidity, and quality governance. In particular, these cases are important for examining whether decentralized architectures deepen exchange while preserving, weakening, or reconfiguring the informational basis of environmental integrity (Cong & He, 2019; Parhamfar et al., 2024).

Industry-integrated models are the most relevant for the production side of the TMP framework. Cases such as Polestar, Volvo Cars/Circular, VeChain-linked applications, and Mercedes-Benz Acentrik are included because they shift the analytical focus from secondary trading toward traceability, data-sharing, and operational visibility in supply chains. Their importance lies less in tokenization itself and more in whether digital infrastructures help make upstream and downstream carbon information more observable, attributable, and potentially contractible (Molinaro et al., 2026; Patro et al., 2025; Zhu et al., 2024).

The comparative analysis then uses pattern matching in the sense of Yin (2018). The aim is not to force each case into a confirmation narrative, but to examine whether the empirical patterns observed across these four clusters support, complicate, or qualify the central transmission claim of the TMP framework. If digital infrastructures improve tradability without improving quality differentiation, then the market-side mechanism remains incomplete. If physically anchored systems improve traceability but remain weakly connected to market incentives, then the production-side mechanism remains underdeveloped. The cases are therefore used not only to illustrate the framework, but also to identify its limits and boundary conditions.

3.3.5. Evidence categories

To avoid relying on one type of source alone, the case analysis distinguishes among three categories of evidence:

- **Technical and architectural evidence, including project documentation, registry materials, and publicly available information on infrastructure design;**
- **Market-operation evidence, including platform disclosures, exchange materials, and, where available, infrastructure-specific transaction records or on-chain activity indicators;**
- **Operational and traceability-related evidence, including corporate disclosures, project-level implementation materials, and third-party assessments where available.**

These sources are not treated as equally probative. Corporate sustainability reports, for example, can indicate implementation priorities or traceability efforts, but they are not treated as independent causal proof of emissions reduction. These sources are not equally probative, and cross-case comparability remains imperfect across infrastructures, jurisdictions, and governance settings. The comparative case analysis should therefore be read as an effort to identify institutional variation and recurrent patterns rather than to impose a uniform evidentiary standard across all projects.

3.3.6. Analytical logic

For each case, the analysis asks three questions:

1. Does the initiative appear to reduce verification frictions or improve traceability in a meaningful way?
2. Does it primarily expand liquidity and tradability, or does it also strengthen quality differentiation and physical anchoring?
3. Does its institutional design appear capable of transmitting digital improvements into stronger production-side incentives?

This structure also leaves room for contrary evidence. The TMP framework would be weakened if digital transparency consistently failed to reduce practical verification frictions, if tokenization deepened liquidity without improving quality differentiation, or if physically integrated systems offered no meaningful traceability advantage over conventional arrangements.

4. Results

4.1. Results from the computational laboratory

The simulation results should be read with some care. They are model-dependent comparative outcomes, not externally valid treatment effects. Within that narrower frame, they remain informative.

Across benchmark runs, the digital-infrastructure scenarios generate a market environment that is more stable than the baseline. When verification frictions remain high, trading tends to be thinner, price formation is more sensitive to quality uncertainty, and execution is more uneven. Once verification-related frictions are reduced, participation increases, execution conditions improve, and average market valuation rises relative to the baseline.

In the benchmark specification, this difference appears as a positive premium in the digital-infrastructure scenario. The figure itself should not be over-interpreted. Its significance is internal to the model. It indicates that, under the specified assumptions, lower verification frictions can change market outcomes in a systematic direction.

More revealing than the premium itself is the conditionality behind it. The model suggests that digital trust, as represented here, does not operate independently of market structure. Its effect is stronger when lower verification frictions are matched by better execution conditions and some improvement in quality differentiation. It is weaker when the market remains constrained by spread-related frictions or persistent uncertainty about asset quality.

The revised liquidity specification reinforces this point. Once execution depends not only on market depth but also on spread-related frictions and order-level price deviation, additional participation no longer guarantees smooth exchange. In some parts of

the parameter space, deeper markets improve execution only modestly. In others, the gains are clearer. This pattern is useful because it prevents the model from implying that more digital participation automatically produces efficient outcomes.

The sensitivity analysis points in the same direction. Across plausible parameter ranges, the basic comparative result remains broadly stable: lower verification frictions tend to improve market functioning relative to the baseline. But the size of the effect changes materially with parameter values. Some scenarios produce only limited gains. Others generate more pronounced ones. That variation is substantively important. It suggests that the mechanism is plausible within the modeled environment, but clearly contingent.

A closer inspection of the sensitivity analysis shows that the main comparative result is directionally stable, but far from invariant in magnitude. When the transparency effect is set near the lower bound of the admissible range, the simulated valuation advantage of the digital-infrastructure scenario becomes noticeably smaller, especially in runs where residual informational noise remains high. A similar attenuation appears when the cost-reduction coefficient is weak, suggesting that lower verification uncertainty alone does not generate substantial market improvements if settlement and reconciliation frictions remain largely intact.

The same pattern appears in the revised execution block. When spread-related frictions are high or when bid-ask deviations remain persistent, the contribution of additional market depth to order matching becomes more limited. Under such conditions, participation increases do not translate mechanically into smoother exchange. By contrast, the comparative gains are more pronounced in parameter regions where verification-related uncertainty falls and execution remains less constrained by spread dispersion and price mismatch.

These sensitivity patterns matter for interpretation. They indicate that the model does not identify a fixed blockchain premium or a universal digital-market effect. What it identifies is a conditional mechanism: under plausible but non-trivial institutional assumptions, lower verification frictions can improve market functioning, yet the scale of that improvement depends on how informational and execution conditions interact. This is precisely why the quantitative outputs of the ABM are interpreted in this paper as scenario-dependent comparative results rather than externally valid estimates. In that sense, the robustness exercise supports the internal coherence of the TMP mechanism while also making clear that its market implications are sensitive to institutional design.

4.2. Comparative case findings

The case analysis yields three broad patterns.

The first is that many digital-carbon initiatives appear better at improving process visibility than at establishing strong evidence of environmental integrity. Several projects clearly aim to improve record consistency, settlement, transferability, or the audit trail around issuance and retirement. These are not trivial gains. In fragmented market settings, they may reduce administrative frictions and improve procedural clarity. Even so, the cases also suggest that better transaction architecture does not by itself settle questions about the quality of the underlying carbon asset.

The second pattern is a distinction between liquidity-oriented models and physically anchored models. Liquidity-oriented systems, including several tokenization and exchange-focused initiatives, concentrate on market access, settlement efficiency, and program-maturity. Their main contribution lies at the market layer. In some settings, they can reduce participation barriers and broaden exchange. But that does not necessarily mean they strengthen the connection between traded claims and independently verifiable emissions reduction.

Physically anchored models perform a different function. These include industrial traceability systems, product-level carbon tracking arrangements, and projects tied more directly to supply-chain information flows. Their value lies less in maximizing secondary-market liquidity and more in making emissions-related information observable, attributable, and in some cases auditable across organizational boundaries. Because of that, they appear more relevant to the production side of the TMP framework.

Recent developments at the verification layer point in the same direction. Verra's approval of the first credits issued under its dMRV pilot is important not because it proves downstream decarbonization effects directly, but because it shows that digitally structured, higher-frequency monitoring is beginning to enter the issuance process itself. That development matters for the present argument. It suggests that the relevant frontier may lie less in tokenization alone than in the gradual integration of digital verification architectures into the institutional core of carbon-credit generation and approval.

The third pattern is that hybrid institutional arrangements often look more credible than purely financial digitalization. This point should not be pushed too far. The available evidence remains too uneven for strong causal claims. Still, across the cases, projects that combine clearer registry discipline, some form of oversight, and more granular data architecture appear better positioned to manage the tension between liquidity and integrity.

At the same time, the cases reveal obvious limits. Many projects remain early-stage. Evidence is often selective. Some claims about traceability or emissions impact rely heavily on self-disclosed materials. Source-level data integrity remains a live issue in nearly every setting. A ledger can preserve records after entry. It does not guarantee that those records were accurate when first produced.

A more concrete illustration of this limit can be seen in the history of tokenized carbon bridging itself. Toucan Protocol is relevant not only because it helped scale on-chain carbon circulation, but also because the controversy surrounding tokenized retired credits exposed a deeper governance tension: market liquidity can expand faster than registry rules and verification semantics adapt. Read in this way, Toucan is not simply a DeFi infrastructure case. It is also a limiting case, showing that digital circulation and environmental integrity may diverge when tokenization outpaces institutional agreement over what exactly is being traded and how those claims should be governed.

Not all cases in the sample point in the same direction, and this asymmetry matters for interpretation. Several tokenization-oriented initiatives appear to improve tradability more clearly than verification quality, while a number of early-stage projects remain too immature to support any firm inference about operational decarbonization effects. In such settings, visible deployment, platform

activity, or market attention should not be confused with demonstrated environmental credibility. This limitation is especially relevant where case materials rely heavily on project documentation, market-facing disclosures, or self-reported sustainability narratives rather than independent longitudinal evaluation. For that reason, the comparative case analysis in this paper is used primarily to map institutional variation and boundary conditions, not to infer uniform success across blockchain-related carbon-market initiatives.

4.3. Bringing the two strands together

Taken together, the simulation and case findings support a cautious version of the TMP framework.

The simulation indicates that if verification frictions fall and execution conditions improve, market outcomes can shift in a more favorable direction. The cases suggest that whether this happens in practice depends heavily on institutional design. Digital infrastructure alone is not enough. Systems focused mainly on tokenization and exchange can improve tradability while leaving asset-quality uncertainty largely unresolved. Systems with stronger physical anchoring appear more promising for operational traceability, but they remain constrained by implementation costs, governance quality, and uneven evidence.

The two strands therefore do not “prove” one another. They speak to different parts of the same problem. The simulation clarifies the internal logic of the mechanism under stylized assumptions. The cases show how that mechanism becomes more fragile and more complicated once it meets real institutions.

5. Discussion

5.1. What the TMP framework adds

The main value of the TMP framework is not that it introduces wholly new variables. Its contribution is more modest, but also more useful. It links three conversations that are often kept apart: blockchain as verification infrastructure, carbon markets as exchange systems, and cleaner production as a response to incentives.

This helps sharpen the real economic question. The issue is not whether blockchain is “good” for carbon markets in some broad sense. The issue is whether digital systems change the informational conditions under which carbon assets are priced and evaluated. If they do, they may reshape parts of market functioning. If those market signals become more credible, they may also matter for production decisions. If not, then the market becomes more digital without becoming more informative in any meaningful environmental sense.

This is also why the term digital trust should be used carefully. In this paper, algorithmic trust does not mean blind confidence in code. It refers more narrowly to a reduction in verification uncertainty under specific technical and institutional conditions. Even then, digital infrastructure does not remove governance. It reallocates where governance pressure falls.

5.2. Why liquidity cannot carry the argument alone

One theme runs through the analysis: liquidity matters, but it cannot carry the whole argument. In carbon markets, deeper trading is only useful if it improves allocation without weakening credibility. A liquid market for low-quality claims is still a low-quality

Table 1

Key parameter settings and variable definitions for the artificial carbon market.

Parameter/Variable	Symbol	Benchmark/Range	Description
Simulation periods	(T)	200	Total number of simulation steps
Number of emitters	(N_E)	100	Firms with heterogeneous compliance or offset needs
Number of verifiers	(N_V)	10	Institutions affecting verification conditions
Number of digital infrastructure providers	(N_D)	5	Providers of blockchain-enabled market infrastructure
Transparency effect	(θ)	0.10 [0.05, 0.20]	Effect of digital verification on perceived asset quality
Cost-reduction coefficient	(β)	0.15 [0.05, 0.30]	Marginal reduction in transaction frictions under digital infrastructure
Depth sensitivity	(α)	0.80 [0.50, 1.20]	Contribution of market depth to order matching
Spread sensitivity	(γ)	1.00 [0.60, 1.40]	Effect of spread-related friction on order matching
Price-gap sensitivity	(λ)	0.75 [0.40, 1.20]	Effect of bid-ask deviation on order matching
Baseline transaction cost	(c_0)	1.00	Normalized benchmark transaction cost
Residual informational noise	$(\varepsilon_{i,t})$	stochastic	Residual uncertainty in perceived asset quality
Carbon price	(P_t)	endogenous	Market-clearing price in each period
Market depth	$(Depth_t)$	endogenous	Aggregate order-book participation
Bid-ask spread	$(Spread_t)$	endogenous	Spread proxy in the artificial market
Bid price	(p_t^b)	endogenous	Representative bid price at time
Ask price	(p_t^a)	endogenous	Representative ask price at time
Latent asset quality	(q_i)	heterogeneous	Intrinsic environmental integrity of asset
Digital infrastructure status	(B_t)	0 or 1	Indicates whether the digital infrastructure condition is activated

Note: This table reports the benchmark values and sensitivity ranges used in the agent-based model. The benchmark values are not presented as econometric estimates. Rather, they are scenario parameters disciplined by stylized institutional evidence, project documentation, and related research on digital verification, settlement, and traceability. Endogenous variables are generated within the model. Parameter ranges are used for robustness and sensitivity analysis rather than formal calibration.

Table 2

Comparative analysis of 16 blockchain-related carbon-market initiatives.

ID	Case name	Governance model	Infrastructure type	Main analytical relevance to the TMP framework	Principal source basis
C1	Project Ensemble (Hong Kong)	State-led	Permissioned ledger	Illustrates sovereign or quasi-sovereign settlement design, tokenized deposits, and delivery-versus-payment architecture for carbon-related transactions	HKMA (2024) and related institutional architecture materials
C2	Climate Impact X (CIX)	Consortium/regulated	Exchange-linked infrastructure	Relevant to trust-oriented market design, higher-integrity marketplace positioning, and exchange-based infrastructure for carbon trading	Exchange rules, platform materials, and market-facing documentation; see also the original draft's use of CIX trading and platform materials
C3	China Green Trust Chain	State-led	National consortium infrastructure	Illustrates data sovereignty, regulatory integration, and the use of digital infrastructure in an expanding national carbon-market setting	Policy and institutional materials as described in the original case table
C4	Zimbabwe Registry (A6 Labs)	State-led	National ledger	Relevant to sovereign registry control, centralized accounting, and attempts to reduce double counting and revenue leakage	Public registry and policy-oriented materials as indicated in the original case table
C5	J.P. Morgan Kinexys	RWA/institutional financial	Private or consortium infrastructure	Relevant to registry-layer tokenization, institutional custody logic, and the financial integration of carbon-related claims	Institutional disclosures and project-level materials as indicated in the original case table
C6	Northern Trust Carbon Ecosystem	RWA/institutional financial	Private ledger	Relevant to lifecycle servicing, custody, and institutionalization of carbon assets as regulated or serviceable financial products	Corporate and platform materials as described in the original case table
C7	ACX (Abu Dhabi)	Regulated exchange	DLT-enabled exchange	Illustrates exchange architecture, securitization logic, and regulated market design in carbon trading	Exchange materials and market-facing public documentation
C8	Thallo	Bridging/market-access model	Polygon/Celo-linked bridging infrastructure	Relevant to liquidity aggregation, bridging infrastructure, and improved market access for fragmented carbon-credit supply	Public platform materials and project documentation
C9	Carbonmark (KlimaDAO)	DeFi-native	Public chain	Illustrates permissionless access, continuous market participation, and secondary-market depth in tokenized carbon trading	Public platform materials and on-chain/API-facing infrastructure descriptions
C10	Senken	DeFi-native	Public chain	Relevant to quality filtering, oracle-based data inputs, and algorithmic screening of carbon-market assets	Platform materials, third-party rating references, and the original draft's mention of BeZero-related audit logic
C11	Toucan Protocol	DeFi-native	Multi-chain bridge	Illustrates tokenization at scale and, at the same time, the tension between rapid on-chain circulation and unresolved verification semantics	Public platform materials, on-chain bridge documentation, and controversy surrounding tokenized retired credits as reflected in the original case logic
C12	Regen Network	DeFi-native/ecological	Layer-1 ecological chain	Relevant to broader ecological asset tokenization and community-oriented verification architecture beyond conventional carbon-credit trading	Project documentation and protocol materials
C13	Polestar (EV)	Industry-integrated	Private traceability infrastructure	Relevant to Scope 3 traceability, upstream raw-material visibility, and operational carbon data in supply chains	Corporate sustainability disclosures, supply-chain traceability materials, and Polestar-related public reporting noted in the original draft
C14	Volvo Cars/Circular	Industry-integrated	Private traceability infrastructure	Relevant to battery passport design, carbon and origin traceability, and the physical anchoring of digital records to supply-chain processes	Volvo Cars public materials, Circular-related traceability documentation, and corporate sustainability disclosures cited in the original draft
C15	VeChain (VeBetterDAO)	Industry-integrated/consumer-facing	Public/consortium-linked model	Relevant to downstream incentive design, behavioral participation, and carbon-related consumer engagement mechanisms	Platform materials and project-level ecosystem documentation
C16	Mercedes-Benz Acentrik	Industry-integrated	Data exchange platform	Relevant to controlled data sharing, supply-chain carbon-data exchange, and data-sovereignty issues in industrial coordination	Corporate/platform materials and public documentation on data-sharing architecture

Note: The cases are organized through theoretical sampling rather than random sampling. Their purpose is to capture variation across governance structure and degree of physical anchoring, following the original draft's four-way grouping into state-led, RWA/institutional financial, DeFi-native, and industry-integrated models. The cases are therefore used to map institutional variation and boundary conditions rather than to constitute a representative sample of all blockchain-related carbon-market initiatives.

Sources and evidentiary basis: In line with the original draft, the comparative assessment draws on three categories of evidence: technical verification materials (for example, smart-contract, white-paper, or protocol documentation), market data (including on-chain explorers and exchange reports, such as PolygonScan for Toucan and exchange materials for Climate Impact X), and operational evidence (including corporate sustainability disclosures such as those used for Volvo and Polestar, and third-party ratings or audit-related materials such as those noted for Senken). These sources are

not treated as equally probative. They are used primarily to establish governance structure, infrastructure type, and stated implementation scope, not to infer uniform causal effects on emissions reduction.

market.

This point is easy to miss in discussions of tokenized environmental assets. Easier access, faster settlement, and wider participation are all attractive features. They may even be necessary in fragmented settings. But they are not sufficient. If tokenization reduces trading frictions faster than it improves verification, then the likely result is not the disappearance of the lemon-market problem. It is its digitization.

5.3. *Physical anchoring and the problem of source-level truth*

The comparative findings suggest that digital systems become more credible when they are more tightly anchored to physical processes. This is where digital MRV, product-level carbon tracking, and supply-chain traceability become especially important.

In principle, physically anchored systems can improve carbon visibility where conventional reporting struggles. Scope 3 emissions are the obvious example. Data are dispersed, often delayed, and difficult to verify across tiers of suppliers. If digital systems make such data easier to trace, they may change not only reporting practice but the incentives surrounding operational decarbonization.

Still, that argument needs restraint. Physical anchoring is not the same as truth. IoT devices can be tampered with. Data entry can be biased. Firms can strategically disclose some information while withholding other parts. Digital MRV can reduce some forms of opacity, but it does not remove the source-data problem.

5.4. *Rethinking hybrid governance*

The paper's discussion of hybrid governance is narrower in the revised version. It does not claim that any one hybrid model is universally optimal. It argues something simpler. Digital carbon markets face two coordination problems at once. One concerns settlement and exchange architecture. The other concerns data generation and verification at the source.

Systems that address only the first problem may improve circulation while leaving quality uncertainty largely intact. Systems that address only the second may improve measurement without connecting that information to broader market coordination. This is why some form of layered governance looks economically plausible.

In practice, that likely means at least three things: a registry or settlement layer that reduces ambiguity and duplicate claims; a traceability layer that links carbon claims to more granular operational information; and an oversight layer that addresses device integrity, liability, auditing, and source-level reporting incentives.

This is why regulated marketplace infrastructures such as Core Climate are analytically useful: their relevance lies less in proving production-side outcomes than in showing how exchange, custody, and settlement can be organized under clearer institutional discipline than in purely permissionless environments.

5.5. *Limitations and future research*

The paper also has limits.

First, the ABM remains stylized. The revised specification is more transparent than the earlier version, especially regarding parameter discipline and execution conditions, but it still relies on simplified behavioral rules. It helps clarify mechanism plausibility. It does not estimate real-world effect size.

Second, the case evidence is heterogeneous and incomplete. Projects vary in maturity and transparency. Failures are harder to observe than visible survivors. Some claims still rely heavily on self-disclosed materials. The case analysis is therefore suggestive rather than definitive.

Third, the paper does not directly measure Green Total Factor Productivity. GTFP remains relevant as a broader lens for thinking about cleaner production, but it is not estimated here. What the paper identifies instead are mechanisms and institutional conditions that may matter for future work using firm-level or panel data.

Broader policy-oriented assessments in the European context make a similar point: the value of blockchain and related digital tools in climate governance depends heavily on institutional fit, data quality, and the surrounding verification regime rather than on ledger adoption alone. This broader policy literature reinforces the need to treat digitalization as an institutional design question rather than a self-sufficient technological solution.

6. Conclusion

This paper examined whether and under what conditions blockchain-based infrastructures may reshape carbon markets and support cleaner production incentives. To do so, it developed a Technology–Market–Production framework linking digital verification architecture, market microstructure, and production-side decarbonization incentives. The analysis combined a mechanism-oriented agent-based model with a comparative assessment of 16 blockchain-related carbon-market and industrial traceability initiatives.

Three conclusions follow.

First, digital infrastructures that reduce verification frictions may improve market functioning, but the effect is conditional rather than automatic. In the computational laboratory, lower verification frictions are associated with more stable exchange outcomes and

higher average market valuation relative to the baseline. But the size of those gains depends on execution conditions, information quality, and parameter choices.

Second, more liquidity does not necessarily mean better environmental performance. Tokenization can make carbon claims easier to trade without resolving uncertainty about the integrity of the underlying asset. Where digitalization improves circulation more than verification, the market may look more efficient while remaining informationally weak.

Third, systems with stronger physical anchoring, especially those tied to digital MRV and supply-chain traceability, appear more relevant to cleaner production than purely exchange-oriented models. Even here, the evidence remains incomplete, and the paper does not claim that current digital systems directly improve GTFP. The narrower conclusion is that digital market reform appears more credible when improved exchange is tied to improved source-level information.

In the end, the future of digital carbon markets depends less on blockchain in the abstract than on the institutional quality of the systems into which it is inserted. The key issue is not whether carbon assets can move faster. It is whether market infrastructure can be improved in ways that preserve a credible link between financial exchange and physical decarbonization. The argument of the paper is therefore not that blockchain offers a stand-alone solution, but that digital infrastructures may matter when they are embedded in a broader institutional architecture capable of governing verification, market coordination, and source-level data integrity together.

Code availability and reproducibility

The simulation code and parameter configuration files used in this study are available from the authors upon reasonable request. To improve reproducibility, the revised manuscript reports the benchmark parameter values, robustness ranges, and the indicative basis for parameter selection in Table 1 and Appendix Table A1. The agent-based model is presented as a scenario-based computational framework rather than as a fully calibrated empirical forecasting tool, and the reported quantitative outputs should be interpreted accordingly (see Table 2).

Appendix A. Parameter rationale and robustness design

The revised study does not treat the ABM parameters as econometrically estimated quantities. Instead, the model uses parameter discipline. The benchmark values and ranges reported below are intended to represent plausible institutional scenarios rather than point estimates derived from a standardized observational dataset. The ranges are informed by public project documentation, stylized evidence on digital verification and settlement infrastructure, and the economic role each parameter plays in the model. Their purpose is to make the scenario logic transparent and to support structured sensitivity analysis.

Appendix Table A1

Parameter rationale and indicative basis

Parameter	Symbol	Benchmark/ Range	Economic meaning	Indicative basis
Transparency effect	(θ)	0.10 [0.05, 0.20]	Measures the extent to which digital verification reduces quality-related uncertainty in carbon assets	Bounded as a moderate scenario parameter reflecting improved traceability and record transparency under digital verification architectures; informed by the logic of blockchain-based auditability and digital MRV discussions rather than direct micro-estimation
Cost-reduction coefficient	(β)	0.15 [0.05, 0.30]	Captures the marginal reduction in reconciliation, settlement, and verification frictions under digital infrastructure	Set as a conservative-to-moderate scenario range reflecting potential reductions in transaction frictions suggested by digital settlement and smart-contract-based infrastructure studies
Depth sensitivity	(α)	0.80 [0.50, 1.20]	Measures how strongly greater market depth improves order matching	Chosen to reflect a positive but non-dominant role for depth in execution; bounded to avoid the unrealistic assumption that depth alone determines market quality
Spread sensitivity	(γ)	1.00 [0.60, 1.40]	Measures the extent to which wider spreads reduce order matching	Introduced to incorporate a standard execution friction into the revised market microstructure block; range chosen to ensure that spread conditions materially constrain matching in some scenarios
Price-gap sensitivity	(λ)	0.75 [0.40, 1.20]	Measures the effect of bid–ask deviation on match probability	Set to capture the decline in execution likelihood when order-level price expectations diverge; bounded to allow heterogeneity in market tightness across simulation runs
Baseline transaction cost	(c_0)	1.00	Normalized pre-digital benchmark transaction cost	Used as a normalization constant to facilitate comparison across baseline and treatment scenarios
Number of emitters	(N_E)	100	Number of firms with heterogeneous compliance or offset needs	Chosen to create sufficient heterogeneity while preserving computational tractability in repeated Monte Carlo runs
Number of verifiers	(N_V)	10	Number of institutions affecting verification conditions	Set to represent a limited but non-trivial verification layer within the artificial market
Number of digital infrastructure providers	(N_D)	5	Number of actors shaping digital verification and settlement conditions	Chosen to introduce institutional heterogeneity without overcomplicating the model environment

(continued on next page)

Appendix Table A1 (continued)

Parameter	Symbol	Benchmark/ Range	Economic meaning	Indicative basis
Simulation periods	(T)	200	Number of time steps in each simulation run	Long enough to observe post-shock adjustment and repeated interaction after the digital-infrastructure condition is introduced

Note: The parameter values reported here are not presented as empirically calibrated estimates. They are scenario parameters disciplined by institutional logic, public project materials, and related research on blockchain-based verification, settlement, and traceability. The benchmark values define the reference specification, while the reported ranges are used in sensitivity analysis to assess whether the qualitative sign of the main comparative results remains stable under plausible alternative assumptions.

Data availability

Data will be made available on request.

References

- Akerlof, G. A. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3), 488–500. <https://doi.org/10.2307/1879431>
- Catalini, C., & Gans, J. S. (2020). Some simple economics of the blockchain. *Communications of the ACM*, 63(7), 80–90. <https://doi.org/10.1145/3359552>
- Coase, R. H. (1960). The problem of social cost. *The Journal of Law and Economics*, 3, 1–44. <https://doi.org/10.1086/466560>
- Cong, L. W., & He, Z. (2019). Blockchain disruption and smart contracts. *Review of Financial Studies*, 32(5), 1754–1797. <https://doi.org/10.1093/rfs/hhz007>
- Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review*, 14(4), 532–550. <https://doi.org/10.2307/258557>
- Hong Kong Monetary Authority. (2024). *Project ensemble: Exploring tokenisation for the future financial market*. Hong Kong Monetary Authority.
- Howell, S. T., Niessner, M., & Yermack, D. (2020). Initial coin offerings: Financing growth with cryptocurrency token sales. *Review of Financial Studies*, 33(9), 3925–3974. <https://doi.org/10.1093/rfs/hhz131>
- Jaffe, A. B., Newell, R. G., & Stavins, R. N. (2005). A tale of two market failures: Technology and environmental policy. *Ecological Economics*, 54(2–3), 164–174. <https://doi.org/10.1016/j.ecolecon.2004.12.027>
- Macinante, J. D. (2017). A conceptual model for networking of carbon markets on distributed ledger technology architecture. *Carbon & Climate Law Review*, 11(3), 243–260. <https://doi.org/10.21552/cclr/2017/3/17>
- Molinaro, M., Orzes, G., & Sarkis, J. (2026). Unpacking circular economy practices and carbon emissions relationships: Co-benefits and legitimacy perspectives. *Business Strategy and the Environment*, 35(1), 125–146. <https://doi.org/10.1002/bse.70157>
- Parhamfar, M., Sadeghkhani, I., & Adeli, A. M. (2024). Towards the net zero carbon future: A review of blockchain-enabled peer-to-peer carbon trading. *Energy Science & Engineering*, 12(3), 1242–1264. <https://doi.org/10.1002/ese3.1697>
- Patro, P. K., Jayaraman, R., Acquaye, A., Salah, K., & Musamih, A. (2025). Blockchain-based solution to enhance carbon footprint traceability, accounting, and offsetting in the passenger aviation industry. *International Journal of Production Research*, 63(22), 8235–8268. <https://doi.org/10.1080/00207543.2024.2441450>
- Schletz, M., Nassiry, D., & Lee, M.-K. (2020). Blockchain and tokenized securities: The potential for green finance. *ADBI Working Paper Series*.
- Stiglitz, J. E. (2002). Information and the change in the paradigm in economics. *The American Economic Review*, 92(3), 460–501. <https://doi.org/10.1257/00028280260136363>
- Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). Sage.
- Zhu, J., Feng, T., Lu, Y., & Jiang, W. (2024). Using blockchain or not? A focal firm's blockchain strategy in the context of carbon emission reduction technology innovation. *Business Strategy and the Environment*, 33(4), 3505–3531. <https://doi.org/10.1002/bse.3664>